Certamen 1 Mecánica Clásica Avanzada

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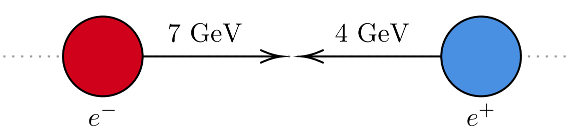
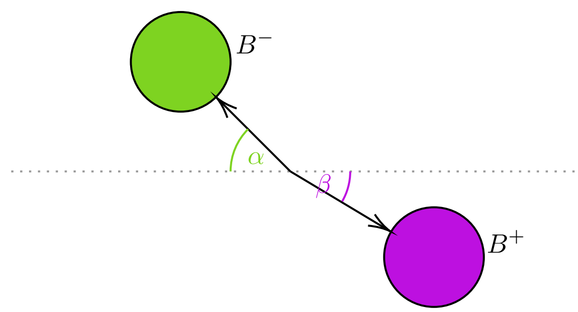
1 Problema 1

The accelerator SuperKEKB collides electrons (e^-) of energy $E_1 = 7\,\text{GeV}$ with positrons (e^+) of energy $E_2 = 4$ GeV. Before the collision, both particles move in opposite directions along the axis z (collision axis in what follows). After the collision, two oppositely charged B-mesons (B^+ and B^-) are produced via the process $e^+\,e^- \to B^+\,B^-$, and one of the B-mesons emerges at angle α_1 with respect to the collision axis. The rest frame masses of both B-mesons are equal to each other, whereas masses of electron and positron are negligibly small.

a) Assuming that one of the mesons emerges at angle α with respect to the collision axis, find the energies and momenta of the produced B-mesons in the laboratory frame. Your result should be expressed as a function of the angle α and energies of the incoming particles.

b) Write out explicit expressions for the Mandelstam variables s, t, u in terms of the energies E_1, E_2 and the angle α in the lab frame.





a) Tomamos $c = 1$. Asumiendo que YZ es el plano de scattering, tenemos

$$\begin{pmatrix} p^\mu_{e^-} \\ p^\mu_{e^+} \\ p^\mu_{B^-} \\ p^\mu_{B^+} \end{pmatrix} = \begin{pmatrix} E_1 & 0 & 0 & |\vec{p}_{e^-}| \\ E_2 & 0 & 0 & -|\vec{p}_{e^+}| \\ \mathcal{E}_1 & 0 & |\vec{p}_{B^-}|\sin\alpha & -|\vec{p}_{B^-}|\cos\alpha \\ \mathcal{E}_2 & 0 & -|\vec{p}_{B^+}|\sin\beta & |\vec{p}_{B^+}|\cos\beta \end{pmatrix} \tag{1.1}$$

De la conservación de 4-momento tendremos:

$$E_1 + E_2 = \mathcal{E}_1 + \mathcal{E}_2 \tag{1.2}$$
$$0 = |\vec{p}_{B^-}| \sin\alpha - |\vec{p}_{B^+}| \sin\beta \tag{1.3}$$
$$|\vec{p}_{e^-}| - |\vec{p}_{e^+}| = -|\vec{p}_{B^-}| \cos\alpha + |\vec{p}_{B^+}| \cos\beta \tag{1.4}$$

De la condición *on-shell*:

$$E_1 \approx |\vec{p}_{e^-}| \tag{1.5}$$
$$E_2 \approx |\vec{p}_{e^+}| \tag{1.6}$$
$$\mathcal{E}_1 = \sqrt{m_B^2 + |\vec{p}_{B^-}|^2} \tag{1.7}$$
$$\mathcal{E}_2 = \sqrt{m_B^2 + |\vec{p}_{B^+}|^2} \tag{1.8}$$

donde m_B es la masa en reposo del bosón B.
De la ec. 1.2 tenemos

$$\mathcal{E}_2 = E_1 + E_2 - \mathcal{E}_1 \tag{1.9}$$

De las condiciones 1.7 y 1.8

$$|\vec{p}_{B^-}|^2 = \mathcal{E}_1^2 - m_B^2 \tag{1.10}$$
$$|\vec{p}_{B^+}|^2 = \mathcal{E}_2^2 - m_B^2 \tag{1.11}$$

Y de las condiciones 1.3 y 1.4:

$$|\vec{p}_{B^+}| \sin\beta = |\vec{p}_{B^-}| \sin\alpha \tag{1.12}$$
$$|\vec{p}_{B^+}| \cos\beta = (|\vec{p}_{e^-}| - |\vec{p}_{e^+}|) + |\vec{p}_{B^-}| \cos\alpha \tag{1.13}$$

Reemplazando las ecs. 1.5 y 1.6 en las ecs. 1.12 y 1.13, elevándolas al cuadrado y sumándolas:

$$|\vec{p}_{B^+}|^2 = |\vec{p}_{B^-}|^2 + 2(E_1 - E_2)|\vec{p}_{B^-}| \cos\alpha + (E_1 - E_2)^2 \tag{1.14}$$

Reemplazando según las ecs. 1.10 y 1.11, y utilizando ec. 1.9:

$$(E_1 + E_2 - \mathcal{E}_1)^2 - m_B^2 = \mathcal{E}_1^2 - m_B^2 + 2(E_1 - E_2)\sqrt{\mathcal{E}_1^2 - m_B^2} \cos\alpha + (E_1 - E_2)^2 \tag{1.15}$$

Usando Mathematica para resolver la ec. 1.15 para $\mathcal{E}_1$ tendremos:

$$\text{FullSimplify}\left[\text{Solve}\left[(E_1 + E_2 - \mathcal{E}_1)^2 - m_B^2 = 2\cos(\alpha)(E_1 - E_2)\sqrt{\mathcal{E}_1^2 - m_B^2} + (E_1 - E_2)^2 - m_B^2 + \mathcal{E}_1^2, \mathcal{E}_1\right]\right]$$

$$\Rightarrow \mathcal{E}_1 = \frac{\sqrt{\cos^2(\alpha)(E_1 - E_2)^2 \left(4E_1^2 E_2^2 + m_B^2 \cos^2(\alpha)(E_1 - E_2)^2 - m_B^2 (E_1 + E_2)^2\right)} - 2E_1 E_2 (E_1 + E_2)}{\cos^2(\alpha)(E_1 - E_2)^2 - (E_1 + E_2)^2} \tag{1.16}$$

Luego, usando la ec. 1.9

$$\Rightarrow \mathcal{E}_2 = E_1 + E_2 - \frac{\sqrt{\cos^2(\alpha)(E_1 - E_2)^2 \left(4E_1^2 E_2^2 + m_B^2 \cos^2(\alpha)(E_1 - E_2)^2 - m_B^2 (E_1 + E_2)^2\right)} - 2E_1 E_2 (E_1 + E_2)}{\cos^2(\alpha)(E_1 - E_2)^2 - (E_1 + E_2)^2} \tag{1.17}$$

b) Tenemos que

$$
\begin{aligned}
s &= \eta(p^\mu_{e^-} + p^\mu_{e^+})(p^\mu_{e^-} + p^\mu_{e^+}) \\
&= (E_1 + E_2)^2 - (E_1 - E_2)^2 \\
&= 4E_1E2 \qquad (1.18) \\
t &= \eta(p^\mu_{B^-} - p^\mu_{e^-})(p^\mu_{B^-} - p^\mu_{e^-}) \\
&= -2E_1\cos(\alpha)\sqrt{\mathcal{E}_1^2 - m_B^2} - 2E_1\mathcal{E}_1 + m_B^2 \qquad (1.19) \\
u &= 0 + 0 + m_B^2 + m_B^2 - s - t \\
&= 2E_1(\mathcal{E}_1 - 2E_2) + 2E_1\cos(\alpha)\sqrt{\mathcal{E}_1^2 - m_B^2} + m_B^2 \qquad (1.20)
\end{aligned}
$$

2 Problema 2

A particle of mass m and charge e scatters in the axially symmetric magnetic field which has only z-component, $\mathbf{B} = B_z(r)\hat{\mathbf{z}}$, where r is the distance from axis z, and in what follows we use cylindrical coordinates (r, φ, z).

a) Write out explicitly the Lagrangian of this system in cylindrical coordinates. Identify its symmetries, cyclic variables and integrals of motion that may exist in this system.

b) Demonstrate that it is possible to integrate the equations of motion and find trajectories $r(t), r(\varphi)$ in general case (for arbitrary $B_z(r)$), only using the integrals of motion.

c) Assume now that the magnetic field vanishes rapidly in the limit $r \to \infty$, so the particles move freely at large distances. A charge particle with initial velocity $\mathbf{v}_0 = v_0\hat{\mathbf{x}}$ scatters in the above-mentioned magnetic field. Find the general solution (analytic expression) which allows to calculate the scattering angle θ as a function of the impact parameter ρ and velocity v_∞ in any magnetic field $B_z(r)$.

   i. Assume that the magnetic vector potential is given by

$$
A_\varphi(r) = \begin{cases} \dfrac{L}{er}\left(1 - \dfrac{r^2}{R^2}\right), & r < R \\ 0, & r > R \end{cases}, \qquad B_z = \begin{cases} -\dfrac{2L}{eR^2}, & r < R \\ 0, & r > R \end{cases}
$$

   Using your general result, find explicitly the scattering angle $\theta(\rho)$ for the angular momentum $M_z = L$.

a) Tenemos la velocidad:

$$
\mathbf{v} = \dot{r}\hat{\mathbf{r}} + r\dot{\varphi}\hat{\varphi} + z\hat{\mathbf{z}} \tag{2.1}
$$

Asumiendo que conocemos $\mathbf{A} = A_\varphi(r)\hat{\varphi}$ tal que $\mathbf{B} = \nabla \times \mathbf{A}$, el lagrangiano tendrá la forma

$$
\begin{aligned}
L &= \frac{1}{2}m\mathbf{v}^2 + e\mathbf{v}\cdot\mathbf{A}(r) \\
&= \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\varphi}^2 + \dot{z}^2\right) + eA_\varphi(r)r\dot{\varphi} \qquad (2.2)
\end{aligned}
$$

De aquí se aprecia que φ y z son coordenadas cíclicas, por lo que tendremos simetría angular y en el eje z:

$$Q_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = mr^2\dot{\varphi} + eA_\varphi(r)r = \text{const.} \tag{2.3}$$

$$Q_z = \frac{\partial L}{\partial \dot{z}} = m\dot{z} = \text{const.} \tag{2.4}$$

Y, ya que el tiempo no aparece explícitamente, se conservará la energía:

$$E = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\varphi}^2 - \dot{z}^2\right) + eA_\varphi(r)r\dot{\varphi} = \text{const.} \tag{2.5}$$

b) De la ec. 2.3, resolviendo para $\dot{\varphi}$:

$$\dot{\varphi} = \frac{Q_\varphi - erA_\varphi(r)}{mr^2} \tag{2.6}$$

$$\Rightarrow \varphi - \varphi_0 = \int_0^t \frac{Q_\varphi - erA_\varphi(r)}{mr^2}\, dt$$

$$\varphi(t) - \varphi_0 = \frac{Q_\varphi - erA_\varphi(r)}{mr^2} t \tag{2.7}$$

Y de la ec. 2.5:

$$\dot{r}^2 = \frac{2}{m}(E - eA_\varphi(r)r\dot{\varphi}) - r^2\dot{\varphi}^2 - \dot{z}^2$$

$$= \frac{1}{m}\left(2E - \frac{Q_z^2}{m} - \frac{(Q_\varphi - erA_\varphi(r))^2}{mr^2}\right)$$

$$\Rightarrow \dot{r} = \sqrt{\frac{1}{m}\left(2E - \frac{Q_z^2}{m} - \frac{(Q_\varphi - erA_\varphi(r))^2}{mr^2}\right)} \tag{2.8}$$

Por lo que

$$\Rightarrow t - t_0 = \int_{r_0}^r \sqrt{\frac{1}{m}\left(2E - \frac{Q_z^2}{m} - \frac{(Q_\varphi - erA_\varphi(r))^2}{mr^2}\right)}\, dr \tag{2.9}$$

Y, para $r(\varphi)$:

$$\frac{dr}{d\varphi} = \frac{\dot{r}}{\dot{\varphi}} = \frac{m\dot{r}r^2}{Q_\varphi - erA_\varphi(r)}$$

$$\Rightarrow \varphi - \varphi_0 = \int_{r_0}^r \frac{Q_\varphi - erA_\varphi(r)}{mr^2} \frac{1}{\sqrt{\frac{1}{m}\left(2E - \frac{Q_z^2}{m} - \frac{(Q_\varphi - erA_\varphi(r))^2}{mr^2}\right)}}\, dr \tag{2.10}$$

En ambos casos tenemos expresiones para $r(t)$ y $r(\varphi)$ que dependen solo de la forma de A_φ y de constantes de movimiento.

c) A una distancia infinitamente grande de la región de potencial tendremos que

$$E = \frac{1}{2}mv_0^2 \tag{2.11}$$

Y, además, como la velocidad inicial se encuentra solo en el eje x:

$$Q_z = m\dot{z} = 0 \tag{2.12}$$

Y, finalmente, en una región alejada donde $A_\varphi = 0$, tenemos el resultado típico:

$$\begin{aligned} Q_\varphi &= m\rho^2\dot{\varphi} + 0 \\ &= mv_0\rho \end{aligned} \tag{2.13}$$

Ahora, tomando la ec. 2.10

$$\varphi - \varphi_0 = \int_{r_0}^{r} \frac{Q_\varphi - erA_\varphi(r)}{mr^2} \frac{1}{\sqrt{\frac{1}{m}\left(2E - \frac{Q_z^2}{m} - \frac{(Q_\varphi - erA_\varphi(r))^2}{mr^2}\right)}}\, dr$$

$$\varphi - \varphi_0 = \int_{r_0}^{r} \frac{mv_0\rho - erA_\varphi(r)}{mr^2} \frac{1}{\sqrt{\frac{1}{m}\left(mv_0^2 - \frac{(mv_0\rho - erA_\varphi(r))^2}{mr^2}\right)}}\, dr$$

$$\varphi - \varphi_0 = \int_{r_0}^{r} \frac{mv_0\rho - erA_\varphi(r)}{r\sqrt{m^2v_0^2r^2 - (mv_0\rho - erA_\varphi(r))^2}}\, dr \tag{2.14}$$

El rango completo de la coordenada angular $\Delta\varphi$ vendrá de integrar en $[r_{\min}, \infty]$, donde $r_{\min}$ es la distancia más cercana al centro del campo, es decir:

$$\Delta\varphi = 2\int_{r_{\min}}^{\infty} \frac{mv_0\rho - erA_\varphi(r)}{r\sqrt{m^2v_0^2r^2 - (mv_0\rho - erA_\varphi(r))^2}}\, dr \tag{2.15}$$

Y, ya que $\theta = \pi - 2\Delta\varphi$

$$\theta(\rho, v_\infty) = \pi - 2\int_{r_{\min}}^{\infty} \frac{mv_0\rho - erA_\varphi(r)}{r\sqrt{m^2v_0^2r^2 - (mv_0\rho - erA_\varphi(r))^2}}\, dr \tag{2.16}$$

i. Considerando

$$A_\varphi(r) = \begin{cases} \dfrac{L}{er}\left(1 - \dfrac{r^2}{R^2}\right), & r < R \\ \\ 0, & r > R \end{cases} \tag{2.17}$$

donde

$$L = Q_\varphi = mv_0\rho \tag{2.18}$$

Tendremos que

$$\theta(\rho, v_\infty) = \pi - 2\int_{r_{\min}}^{R} \frac{L - erA_\varphi(r)}{r\sqrt{(mv_0r)^2 - (L - erA_\varphi(r))^2}}\, dr - 2\int_{R}^{\infty} \frac{L}{r\sqrt{(mv_0r)^2 - L^2}}\, dr \tag{2.19}$$

La primera integral, usando Mathematica:

$$\begin{aligned}
&\int_{r_{\min}}^{R} \frac{mv_0\rho - erA_\varphi(r)}{r\sqrt{m^2v_0^2r^2 - (mv_0\rho - erA_\varphi(r))^2}}\, dr \\
=&L\int_{r_{\min}}^{R} \frac{\left(1 - \dfrac{r^2}{R^2}\right)}{r\sqrt{m^2v_0^2r^2 - \left(1 - \dfrac{r^2}{R^2}\right)^2}}\, dr \\
=&\arcsin\left(\frac{mv_0r}{RL}\right)
\end{aligned} \tag{2.20}$$

La segunda integral, usando Mathematica y `FullSimplify`:

$$\int_{R}^{\infty} \frac{mv_0\rho}{r\sqrt{(mv_0r)^2 - (mv_0\rho)^2}}\, dr = -\arcsin\left(\frac{\rho}{R}\right) \tag{2.21}$$

Finalmente,

$$\theta(\rho, v_\infty) = \pi - 2\arcsin\left(\frac{mv_0r}{RL}\right) + 2\arcsin\left(\frac{\rho}{R}\right) \tag{2.22}$$